



The effects of high school personal financial education policies on financial behavior[☆]

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ABSTRACT

High schools in the United States are increasingly requiring their students to complete financial education prior to graduation. This study estimates the effects of these requirements on the credit report outcomes of 18- through 21-year-olds—young people just establishing their financial independence. We find that financial education requirements are associated with fewer defaults and higher credit scores among young adults, but this general finding masks important heterogeneity at the state level. We conclude that well-funded teacher preparation may be key to successfully implementing financial education programs.

1. Introduction

As of 2015, nearly half of states in the United States included some form of a personal finance course as part of their high school curriculum. The public policy motivation for school-based financial education requirements is typically to improve the financial behavior of individuals. While paternalistic, there may be merit to this policy rationale. The naiveté of young people may undermine the assumptions of revealed preference models: young people may lack the self-control and awareness to invest in financial knowledge and underestimate the impact of their current financial decisions later in life (Grifoni & Messy, 2012; Lusardi & Mitchell, 2014). State policies reflect this motivation. For example, Colorado's legislation specifies the need for: "knowledge of personal finances that is sufficient to enable a person to manage savings, investment, and checking accounts, to design and maintain a budget, to manage personal debt, to understand consumer credit and finance, to manage personal credit options" (CO Rev. Stat. Ann. (State of Colorado, 2013), Section 22-2-127). In South Carolina, the Financial Literacy Instruction Act (SC Code Ann. (South Carolina Legislature, 2005) Section 59-29-410) includes the preamble: "so students have reduced risk for financial failure after high school." However, new courses could crowd out other learning objectives

in resource-constrained schools. The efficacy of these public investments in financial education need to be understood in order to fully evaluate the costs and benefits of these policies.

These policies reflect a belief that teaching about personal finance, and credit management in particular, at a young age may help prevent future financial problems, such as defaulting on a loan. Younger people today are active in consumer credit markets, and start to develop credit behaviors early in life (Lachance, 2012). Patterns of credit management appear to be persistent over time; young people who have problems with consumer debt also tend to have further problems as they age (Jiang & Dunn, 2013). Moreover, young people are managing more credit than in prior generations, increasing their risks of financial problems later in life (Dwyer, McCloud, & Hodson, 2011; Houle, 2014). These trends place more burden on young people to develop credit management skills earlier in life to successfully handle debt repayments (Campbell & Hercowitz, 2009; Sironi & Furstenberg, 2012). Since 18-year-olds are just taking their first steps to financial independence, the period immediately after high school graduation may be a critical window to help young people to shift to a positive credit trajectory for the long run. Those young people without education may be more likely to learn about managing finances by trial-and-error, a process that could also be

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costly if early mistakes lead to poor credit histories and high borrowing costs later in life (Campbell & Hercowitz, 2009; Jiang & Dunn, 2013).

However, several review studies conclude that financial education has limited or no ability to influence future financial behaviors (Fernandes, Lynch Jr, & Netemeyer, 2014; Hastings, Madrian, & Skimmyhorn, 2013). Other studies find improvements in measured financial knowledge, based on post-course assessments, but do not measure behaviors after leaving school (Hospido, Villanueva, & Zamarro, 2015; Walstad, Rebeck, & MacDonald, 2010; Walstad, Tharayil, & Wagner, 2016). At least one study finds negative effects of a school-based financial education on some behaviors. In a randomized trial in Brazil, where financial education included three semesters of high school coursework, teacher training, interactive learning, and take-home assignments that involved parents, Bruhn, de Souza Leao, Legovini, Marchetti, and Zia (2016) found improvements in student financial knowledge, as well as positive behaviors such as savings. However, they also found that students in the program were *more* likely to use high-cost credit.

In the US, there are only a few studies of the effects of high school financial education policies on behaviors after graduation. The first study, by Bernheim, Garrett, and Maki (2001), found positive effects of 14 state financial education policies on savings by middle-ages. Cole, Paulson, and Shastry (2015), following the classification used by Bernheim et al. (2001), find no effects of state financial education policies on credit scores, debt balances, or foreclosures at ages 46 to 65, however. Brown, Grigsby, van der Klaauw, Wen, and Zafar (2016) identify 17 states where financial education policies were reported to be in place in high schools between 1998 and 2012, and estimate the effects of these policies on debt levels and loan defaults among 19- to 29-year-olds. That analysis shows that states defined as having financial education policies are associated with reductions in non-student debt and rates of loan defaults. However, the authors also find that economics education, which is estimated separately from financial education, is associated with young people borrowing and defaulting more. Since personal finance and economics are often taught as part of the same course, this finding is difficult to interpret. Collectively, these findings do not provide educators or policymakers a clear prescription for influencing the financial behaviors of young people.

To address this gap, we study two states that enacted financial education graduation mandates in 2007.¹ We use a difference-in-differences method and quarterly credit report data for people from age 18 through age 21 to estimate the relative effects of each state's policy. Comparing student graduation-age cohorts in these states to students in demographically similar states with no financial education policies, with control states defined using synthetic control methods, we show improvements in credit behaviors among students exposed to financial education policies.

This study advances our understanding of financial education policies in four important ways. First, we conduct an expansive data collection exercise in which we document the specifics of each state's financial education policy and its implementation into schools. We identify a subset of states that require financial education courses, or require content to be integrated in existing required courses—often economics—before students can graduate from high school.

Second, we document heterogeneity that goes beyond variation in course requirements. We focus on two states enforcing financial education requirements post-2000 that were not coincident with changes in economic or math education reforms. Even with similar graduation mandates implemented in the same year, there is heterogeneity in implementation in these two states, and this leads to differing credit

outcomes for young people. This is the first study to investigate the heterogeneity of high school financial education policies on financial behaviors at young ages.²

Third, we document the dates states approved legislation or administrative orders related to its financial education policy, as well as the date of implementation in schools. We show using the policy approval date, rather than the actual year of implementation, produced null estimates of the effects of financial education—which at least partially explains why prior studies show conflicting results. We further show there heterogeneity within states over time after policies are in place; each successive graduating class under a new policy shows increasing improvements on credit behaviors, consistent with schools enhancing financial education courses over time.

Fourth, we show the effects of financial education on the credit choices that young people make at ages right after they graduate from high school. While Brown et al. (2016) and Cole et al. (2015) examine credit behaviors later in life, financial decisions are more complex as individuals form households and make joint choices about saving and borrowing with partners and spouses.³ It is important to bound the period of observation to compare people within the same age cohort exposed and not exposed to financial education. The behavior of people age 18 through 21—right after high school as they establish independence—is focal period in young adults becoming economically self sufficient. We consider credit outcomes to be especially relevant for this population, and plausibly influenced by financial knowledge. This is an enhancement over prior studies with more dispersed financial outcome measures.

We find that high school financial education graduation requirements, on average, lead to higher credit scores and reduced rates of credit delinquencies, though they do not change the fraction of young adults with credit files. While these estimates are positive for the two states policies we examine, there are different magnitudes of effects even among similar financial education policies. These findings illustrate the challenges of finding average treatment effects of policies across states, and the importance of documenting variations in implementation when using state-level indicators in policy analysis.

2. Documenting state financial education mandates

Both Brown et al. (2016) and Cole et al. (2015) define states with financial education policies using bi-annual reports from the National Council on Economic Education (NCEE). Yet, reporting of policies by even well-informed individuals is challenging. Those in states that have passed policies that have not yet been implemented in schools may respond that their states have a financial education policy, when those programs are still in development and not yet impacting students. For example, New Jersey approved a financial education mandate in 2009. That program was piloted in 2011, and then required for graduation in 2014. However, New Jersey is listed in some reports as having a graduation mandate prior to 2014. This discrepancy highlights one of the problems with studies of state financial education policies; there is no well-defined, validated dataset that documents each state's policy implementation. Moreover, existing reports do not systematically document differences across state policies.⁴

For each state, from 1980 to 2014, we determined whether a personal finance course was required prior to graduation, required to be

¹ A previous version of this paper also included Idaho, however, after consulting with local experts we determined that personal finance had already been included in state standards with the inception of an economics requirement in 2001, as well as directly in Idaho's Math Common Core requirements. This removed it from our inclusion criteria. The earlier version can be found at <https://www.federalreserve.gov/pubs/feds/2014/201468/201468pap.pdf>.

² Tennyson and Nguyen (2001) show heterogeneity in the effects of financial education policies across states on financial knowledge measured in standardized assessments, but not on behavior.

³ Note that all of these studies assume people currently reside in the state in which they completed high school; the validity of this assumption decreases when older individuals are included in the sample.

⁴ For example, Cole et al. (2015) concede that data issues are a challenge for their analysis: "We have been unable to obtain data from state boards of education that would allow us to include mandates after 1982" (p. 665).

Table 1
State personal finance graduation requirements by year or implementation.

State	Grad Yr	Mandate Yr	Course (credits)	Notes
RIGOROUS: PERSONAL FINANCE GRADUATION REQUIREMENT				
Arkansas	2005	2005	Econ (0.5)	Rev 2009, 2014
Colorado	2009	2008	Econ or Math (0.5)	
Georgia	2007	2004	Econ (0.5)	
Iowa	2011	2009	21st Cent. Skills (**)	
Missouri	2010	2006	PF or Other (0.5)	
North Carolina	2005	2005	Econ (0.5)	
South Carolina	2009	2005	Econ (0.5)	
Tennessee	2011	2006	PF (0.5)	
Texas	2007	2004	Econ (0.5)	
Utah	2008	2003	PF (0.5)	
LESS RIGOROUS: OPTIONAL POLICIES OR MINIMAL STANDARDS				
Nebraska	2011		Career Ed	Required to offer
New Mexico	2003		PF (Bus Ed 2003-12)	Required to offer
South Dakota	2006		Choose PF or Econ	
Wyoming	2002		Soc Stud (1 standard)	
COMPLEX IMPLEMENTATION				
Arizona	various		Econ	Local control
Connecticut	various		Econ	Local control
Idaho	2001 & 2013	2008	Econ (0.5)	
Kansas	2012		Econ (0.5)	Early pilots
Louisiana	2005		Free Enterprise	Natural disaster shock
New Jersey	2014		PF or Econ (2.5)	2011 pilot
Oregon	2013		Soc Stud	Early pilots
Virginia	2008			Local control
West Virginia	various			Local control
NO PERSONAL FINANCE GRADUATION REQUIREMENT OR REQUIREMENT TO OFFER COURSE				
Alabama ⁺ , Alaska, California, Delaware, Florida ⁺ , Hawaii, Indiana, Kentucky, Maine, Maryland, Massachusetts, Minnesota ⁺ , Mississippi, Montana, Nevada, North Dakota, Ohio ⁺ , Oklahoma, Pennsylvania, Rhode Island, Vermont, Washington, Wisconsin				
⁺ indicates policy implemented after 2011				
STATES WITH PRE-2000 EXISTING MANDATES				
Illinois (1968), Michigan (1998), New Hampshire (1993), New York (1996)				

Notes: These policy descriptions end with 2013 for the purpose of our design. Credits in terms of Carnegie units (except New Jersey). ** Iowa defines credits as proficiency on standards and competencies rather than Carnegie Units or time. For full data, see <http://www.montana.edu/urban/financial-edu-database.html>.

offered in schools, or not required. We began with two sources: (1) legislation, where the details of the requirements are included in the text of a legislative act; and, (2) administrative rulings, where public documents describe proposed changes. Next, we contacted each state department of education to request current and past graduation standards. These documents allowed us to verify the graduating classes impacted by each policy. We cross-referenced these records with home schooling guidelines for students by graduation year; these were almost always consistent with the graduation requirements.⁵

A summary of our classification is in Table 1. Data on policies for all 50 states show 15 states with clearly definable changes in financial

education policies between 2001 and 2014. Of these 15 states, 11 states have the most rigorous policies that require personal finance content be completed prior to high school graduation as of a specific graduation year. In these states, the required content is delivered via a new standalone courses, or content incorporated into an existing required course. Four states have state financial education policies that we define as being less rigorous. In these states, schools must offer a personal finance course, or personal finance content as part of other courses, but it is not required to graduate from high school. Four additional states had requirements in place before 2000. Nine states are excluded from our analysis due to confounds; four states allow schools to control curriculum locally, making the identification of policy implementation too difficult. Two states implemented incremental pilots in some districts, one state had personal finance in required Economic standards and implemented a personal finance mandate later in addition, and one state's mandate was part of broader changes in education policies. Louisiana implemented its personal finance course requirement at the same time that Hurricane Katrina disrupted school systems and student populations in much of the state.

Defining financial education policies in this way generates different policy variables than prior studies. Our state financial education classification agrees with the identification used by Brown et al. (2016) for only 13 of the 27 states we define as having state financial education policies (Appendix Table A.1). Our classification includes 13 states labeled as not having a policy in Brown et al. (2016). There are five states (Brown et al., 2016) defined as having policies that our classification does not define as such. Even among the 13 states where both study classifications define a state as having a policy, we differ on the year of implementation in many of states.

We next focus on finding states among 10 with graduation requirement policies we can use to test the effects of financial education. First, we targeted states with curriculum changes occurring after 2005 but before 2009. We select this period to provide sufficient pre-policy periods to establish the parallel trends, as well as three post-periods to track outcomes. Second, we focused on states with no other changes in math and economics education requirements simultaneous to the financial education requirements. Third, we omitted the states that implemented the requirements in the same year of the mandate, since it does not seem plausible to implement a well-developed course as a graduation requirement for students in the same year a mandate is passed. These criteria left just two states: Georgia and Texas. Both implemented personal finance as a mandatory element within a one-semester economics requirement in 2007 and had no other major curricular changes in that year. Each of these states had well-documented curricula and graduation requirements, while integrating personal finance requirements into a course for students in 11th or 12th grade. However, other aspects of these state policies still proved to be wide ranging, as illustrated below.

2.1. Georgia

The Georgia Board of Education first approved a statewide financial education mandate in 2004; the first students affected by the policy graduated in 2007. Prior to 2007, a one-semester economics course was offered in Georgia high schools, but the course was not required to address personal finance topics and there was little conformity in content across schools. Within that course, the Georgia Council for Economic Education (GCEE) developed a personal finance unit, titled "Let's Make it Personal," beginning with the graduating class of 2007. Topics in the unit included financial planning, savings, insurance, and credit. The unit included economic simulations to encourage students to engage in applied learning (Hinojosa et al., 2007; 2009; Walstad & Buckles, 2008).

Thus, Georgia's policy included systematic implementation of standardized course content across schools, as well as student performance testing on personal finance topics. About one-fifth of the course

⁵ When we compared our policy determinations to compilations developed by the National Conference of State Legislatures, data from the NCEE's Survey of the States, reports from the Jump\$tart Coalition for Personal Financial Literacy, and the Center for Financial Literacy at Champlain College. We note that in nearly a third of cases, our classification did not match past reports from these sources.

was focused on personal finance, with the remainder on microeconomics, macroeconomics, and international economics. Standards for personal finance included: (1) Explain that banks and other financial institutions are businesses that channel funds from savers to investors; (2) evaluate the costs and benefits of using credit, including comparing interest rates on loans and credit cards, listing factors that affect credit-worthiness, and explaining the difference between simple and complex interest rates; (3) describe how insurance and other risk-management strategies protect against financial loss; (4) describe how the worker's earnings are determined in the marketplace.

Instructors were provided with a framework to evaluate student knowledge on each of the required areas. The standards document provides sources for assessment options and resources for teaching materials. Teachers were provided specific, interactive tools, such as, "In this lesson students can explore or help in developing a plan for building personal wealth. Students can also get an overview of personal finance strategies. Using the Dallas Federal Reserve's interactive website (www.dallasfed.org/ca/wealth/index.cfm), students can calculate various interest rates and their associated costs."

There are three unique components of Georgia's implementation. First, the state provided funding for two experts to train teachers on the new curriculum. This standardized the course offering and prepared teachers to cover the new personal finance material. After that first year, training was available from GCEE through webinars and with local training organizations. This is an ongoing effort: the GCEE reported training 2,400 teachers in 2015 alone. Second, the state department of education tied school funding to the teacher of this course either being a certified economics teacher or having obtained a broad field certification in social studies, reducing the probability that a teacher with an unrelated specialization (e.g., art or foreign language) was teaching personal finance. Third, student standardized testing included topics from the economics course, including personal finance content.

2.2. Texas

A 2004 amendment of the Texas Education Code (Section 1A-28-28.0021) required inclusion of personal finance materials in a half-credit economics class offered in 9th through 12th grade. Schools were required to implement standardized materials beginning with the graduating class of 2007. In 2006, the State Board of Education approved a list of materials prepared by the Texas Council on Economic Education (TCEE) that merged traditional education and experiential learning in the instruction of personal financial literacy topics. Texas standards are strict; schools can be audited at any time and must demonstrate that textbooks and materials covering the required standards are available to all students. Further, teachers must meet either the economics certification standards or the "broad certification" standards in social studies to teach the course, though the level of enforcement of the certification is unclear.

Unlike standards in Georgia, the Texas Board of Education does not specify the amount of time to be set aside for each topic. Thus, it is not clear if the emphasis of the semester-long course is economics or personal finance. However, the new mandate did provide structure to the economics requirement, by prescribing topics within the approved course content: (1) understanding interest, and avoiding and eliminating credit card debt; (2) understanding the rights and responsibilities of renting or buying a home; (3) managing money to make the transition from renting a home to home ownership; (4) starting a small business; (5) being a prudent investor in the stock market and using other investment options; (6) beginning a savings program and planning for retirement; (7) declaring bankruptcy; (8) understanding the types of bank accounts available to consumers and the benefits of maintaining a bank account; (9) balancing a checkbook; (10) understanding the types of loans available to consumers and how to be a low-risk borrower; (11) understanding insurance; (12) engaging in charitable giving.

Though Texas high school students must pass standardized tests to graduate, there is no information on personal finance on the standardized tests. There are no requirements for teacher training in Texas, though there are four low-cost avenues through which teachers can acquire skills to teach the economics and personal finance course. First, the TCEE and the Federal Reserve Bank of Dallas provide materials on both financial literacy and economics and train teachers directly through conferences. Second, there are 20 Educational Service Centers across the state that train teachers; these are particularly valuable for those living in more rural areas. Third, volunteer-based groups across the state supplement the content provided by the TCEE and the Dallas Fed. Fourth, local credit unions and financial institutions provide opportunities for teachers to obtain additional training. Some of the teacher training, especially through the TCEE, has been funded through grants from the Texas Financial Education Endowment. This fund provides private support for financial capability education and financial literacy in Texas.⁶

3. Data

To estimate the effects of state education policies on credit behavior, we use the Federal Reserve Bank of New York/Equifax Consumer Credit Panel (CCP) to track individual young people. These data are a 5 percent random sample from credit report data records of the credit reporting agency, Equifax. The 5 percent random sample is supplemented with the credit report data for all persons who reside at the same address as the primary individual, yielding a total sample of approximately 40 million credit files each quarter. This panel of individuals is updated to add new credit files and remove the files of deceased persons or those with inactive credit files. Lee and Van der Klaauw (2010) provide a detailed description of the CCP sample design, as well as comparisons of CCP estimates of outstanding credit to other aggregate national estimates. The CCP appears to be a highly valid measure of credit trends at the individual level.

CCP data are based on individual credit records, not households. Most people enter the CCP by opening credit accounts in their own name, although young people can establish a credit history by being an authorized user on another person's account (such as a parent) or having an account cosigned by someone with established credit. They may also enter the data by having a negative record, even if they do not seek out or apply for credit. For example, being delinquent on a non-credit-related bill might result in a derogatory collections record in the credit report. In addition to accounts and collections, young people may enter the credit record because they are listed as having derogatory public records, such as civil judgments, liens for taxes or public fines, or bankruptcy. While 65 percent of 18-to 19-year-olds lack a credit history, only 20 percent of those ages 20 to 24 lack a credit history (Brevoort, Grimm, & Kambara, 2016). We document that there is no systematic bias in terms of which young people are in the CCP for our analysis in the next section.

This analysis estimates the effects of financial education policies on credit behaviors observed in the CCP: delinquencies and credit scores. We use two measures of delinquency. Our first measure is a payment that is 30-days-late, a relatively short-term delinquency that could signal inattention to timely payments for existing accounts. This measure also includes individuals without any open accounts coded as not delinquent, since individuals could enter the data from a collections action. The rate is therefore a credit report with any 30 day late payment relative to the entire population in the CCP that quarter.

Our second measure of delinquency is being 90 or more days behind, a more severe form of delinquency. For an existing credit account, this signals some hardship where individuals cannot or will not make payments. Like the 30-day measure, we also label as not being 90 days

⁶ For example, in the 2016-17 school year, the TFEE fund allocated \$32,000.

delinquent any individuals without any credit accounts. To the extent this education provides information about the costs and benefits of paying bills and obligations on time, we would predict that both 30-day and 90-day delinquencies would be lower among students exposed to financial education. These improved payment records will result in higher credit scores, although the effect may be limited since on-time payments are not the only factor influencing credit scores (Avery, Calem, & Canner, 2004).

Our next outcome measure is the credit score. The CCP has an Equifax Risk Score, which is similar to the FICO score. These scores range from 280 to 850, with a higher score indicative of the borrower being a better credit risk. A credit score is a summary measure of credit behavior (Arya, Eckel, & Wichman, 2011) that has been used in prior studies as an outcome measure for evaluations of financial education programs (Birkenmaier, Curley, & Kelly, 2012). However, credit scores are not available for all people with credit records; missing information or sparse credit histories may make it impossible to calculate a credit score. In order to be scorable, a credit record must have at least one active credit account open for at least 6 months (Brevoort et al., 2016). Especially among young people with few accounts or low credit use, scores may tend to be missing or be relatively low. Missed payments are the most common reason for low credit scores, which suggests scores will be correlated with 30- and 90-day-delinquencies.

People who are age 18 in the credit record are included in the sample each quarter, until they turn age 22.⁷ This four-year moving window from age 18 through age 21 follows for each subsequent cohort of high school graduates, as shown in Fig. 2. The data are organized quarterly from 2001 through 2014, although each individual is included for a maximum of 16 quarters. The four-year window allows us to track people through these critical years as they initially establish credit. This four-year graduation-year cohort also allows us to measure the effects of policies implemented as recently as 2010, allowing enough time after graduation to track credit outcomes.

Notably, this study period could be influenced by the Credit Card Accountability Responsibility and Disclosure Act, or CARD Act, which went into effect in February 2010 (see Debbaut, Kudryak, & Ghent (2014)). Historically, young people enter into the credit market via revolving credit card accounts (Fry, 2013; Sotiropoulos & d'Astous, 2012). The law required credit card applicants younger than age 21 to have a source of income or an adult cosigner before obtaining a card. This law was national in scope, so people in all states were affected. The graduating class of 2007 would not have been prevented from obtaining a credit card, and the class of 2008 would have only been affected by one year or less. The class of 2009, however, would have had more barriers to getting credit, although not when they initially graduated from high school (most would have been 20 when the CARD Act was implemented).

4. Methods

To estimate the effect of each specific financial education requirement on credit behaviors, we use a synthetic control strategy (Abadie, Diamond, & Hainmueller, 2010; Abadie & Gardeazabal, 2003; Hinrichs, 2012) to isolate control states for each treated state. Using state characteristics in 2000, we construct a synthetic control sample for each treated state among the states that did not implement any financial education policies before 2011. The state-level variables used to select and weight the synthetic control states included state gross domestic product, median household income, poverty rate, housing prices, unemployment rate, percent with less than a high school degree, percent graduated from high school, percent graduated from college, percent with some college, Census region and division, percent of private schools, race and ethnic composition, expenditures per pupil, and total

schooling expenditures.⁸ Fig. 1 shows the states in this matched sample with the estimated weights for each control state.

These two states implemented personal finance education with the graduating class of 2007. Instead of using synthetic controls to compare aggregate differences across states in the post-period, we preserve our individual-level data and employ a difference-in-difference specification to compare 18- through 21-year-olds within each treated state with a synthetic control state before and after the graduating class of 2007.

Table 2 shows descriptive statistics for the treatment and synthetic control samples for the graduating class of 2001, followed until 2005. Each state and the synthetic control means for this group, all of whom graduated before any financial education took effect, are substantively similar. Importantly, the first year of entry is similar across states (approximately age 18 and 11 months, or just before the person's 19th birthday). Fig. 3 uses the synthetic control weights to show trends for each outcome. In each state, the parallel trends assumption appears to hold before the education policy is implemented. Fig. 4 shows the same plots for delinquencies.

We use the weights from the synthetic controls to estimate a regression and estimate Eq. (1). Standard errors are clustered at the state level in this specification to account for the policy variation.⁹

$$Y_{ist} = \alpha_0 + \beta_1(T_s \times P1_{it}) + \beta_2(T_s \times P2_{it}) + \beta_3(T_s \times P3_{it}) + \gamma_1 u_{it} + \kappa X_{st} + \delta_s + \eta_t + \epsilon_{ist} \quad (1)$$

We focus on three main outcomes of interest, labeled Y_{ist} : the individual's 30-day delinquencies, 90+ -day delinquencies, and credit score in each quarter, t . T_s is a dummy variable that equals 1 if the individual lived in the treated state in the sample (Georgia or Texas) versus a synthetic control state. $P1_{it}$ through $P3_{it}$ equal 1 if the individual turned 18 in the first, second, and third year the financial education requirement was enacted, or 2007, 2008, and 2009, respectively. These coefficients estimate the effects of the state mandates for each subsequent graduating class from 2007 through 2009 on financial behaviors. u_{it} is the unemployment rate in the young adult's current county of residence. X_{st} includes the median state income in the given year, state support for higher education by year, and state education spending per capita by year. These variables are included to account for changes in state education correlated with the timing of the education. Finally, δ_s represents state fixed effects and η_t represents quarter-by-year fixed effects. As a robustness check, we include state-by-year and quarter fixed effects instead of state and quarter-by-year fixed effects in some specifications.

5. Results

This difference-in-differences estimation is based on three assumptions. First, individuals begin their credit records in the same state where they attended high school; Brown et al. (2016) show that 93 percent of individuals stay in the same state from ages 18 to 22.¹⁰ Second, students graduating from treated states were exposed to the financial education graduation requirement if they turned 18 in or after the year the policy was implemented.¹¹ However, there is no reason to suspect this measurement error varies across states with and without financial education policies, attenuating our estimates. We cannot rule out that individuals in states without financial education policies received the education anyway, as schools can still offer classes on their

⁸ If we exclude state economic variables our results remain consistent.

⁹ Since there are few clusters, we use wild cluster bootstraps (Cameron, Gelbach, & Miller, 2008).

¹⁰ Those that move will contribute to classical measurement error, attenuating the effect, as moving should be uncorrelated with the timing and state implementation of financial education.

¹¹ For example, someone who turned 18 in July of the first mandate year may have started kindergarten with the subsequent cohort. That individual may have a credit file but would not be exposed to treatment until the subsequent year.

⁷ Less than 1% of the sample enters and then exits the data; we omit these individuals.

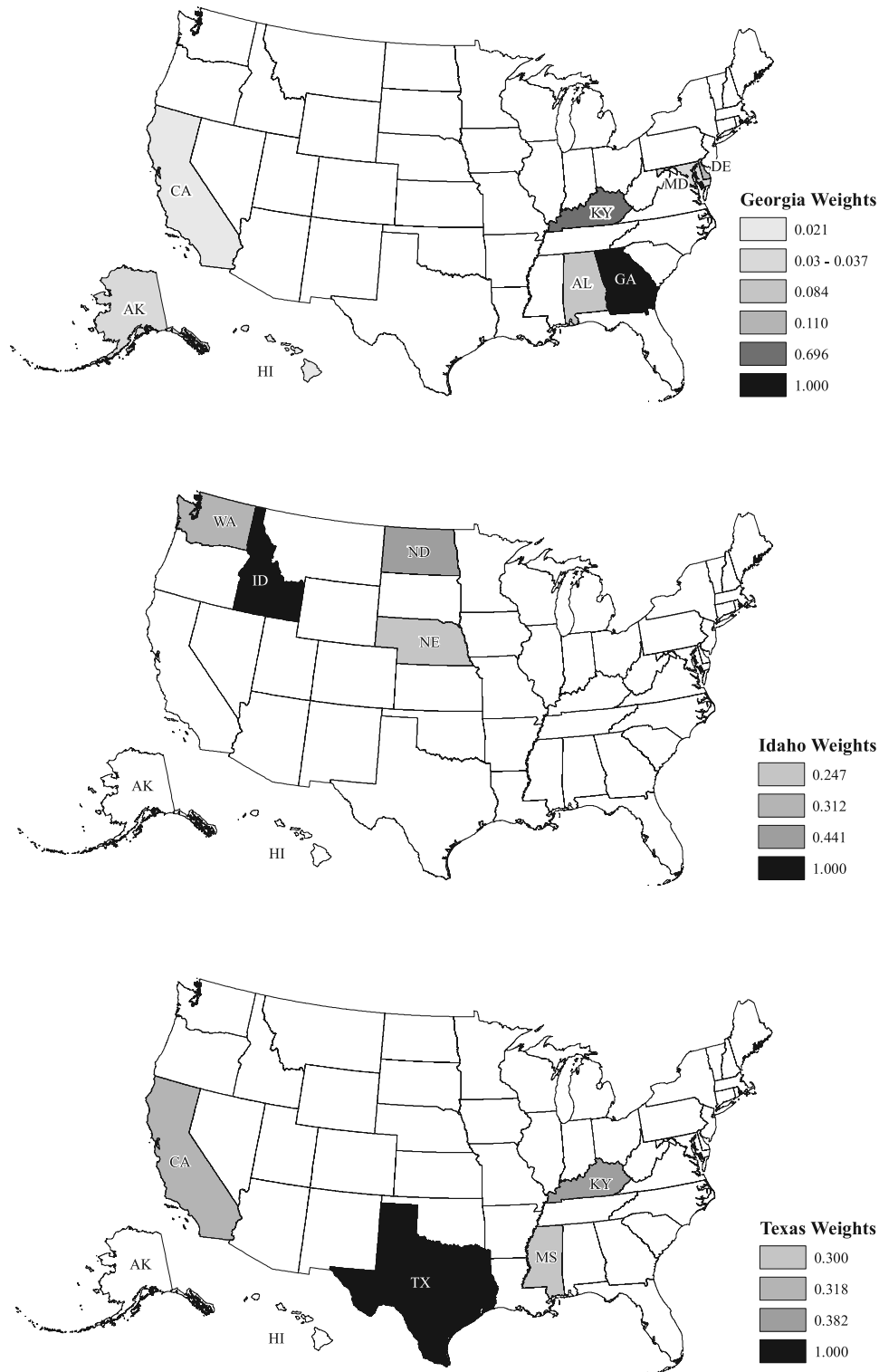


Fig. 1. Treatment and synthetic control states. Notes: Each synthetic control sample was constructed using the following 2000 state-level characteristics: GDP, median household income, poverty rate, HPI, unemployment, education levels, region, division, percent of private schools, expenditure per pupil, race and ethnicity, and total expenditures. Georgia is comprised of AK (0.03), AL (0.084), CA (0.021), DE (0.11), HI (0.021), KY (0.696), MD (0.037); Texas is comprised of CA (0.318), KY (0.382), MS (0.300).

own. Having more young adults who receive education in states without policies will bias our estimates toward zero. The third assumption is that in the absence of policies, individuals in states with financial education policies would have had similar relative trends in credit outcomes to those in control states.

We begin by testing the effect of the date when states announced their policies in [Table 3](#). Prior studies on financial education policies have generally used the date that the state policy was approved to estimate the effects of a new policy. In reality, the date when students actually receive any required financial education might be several years

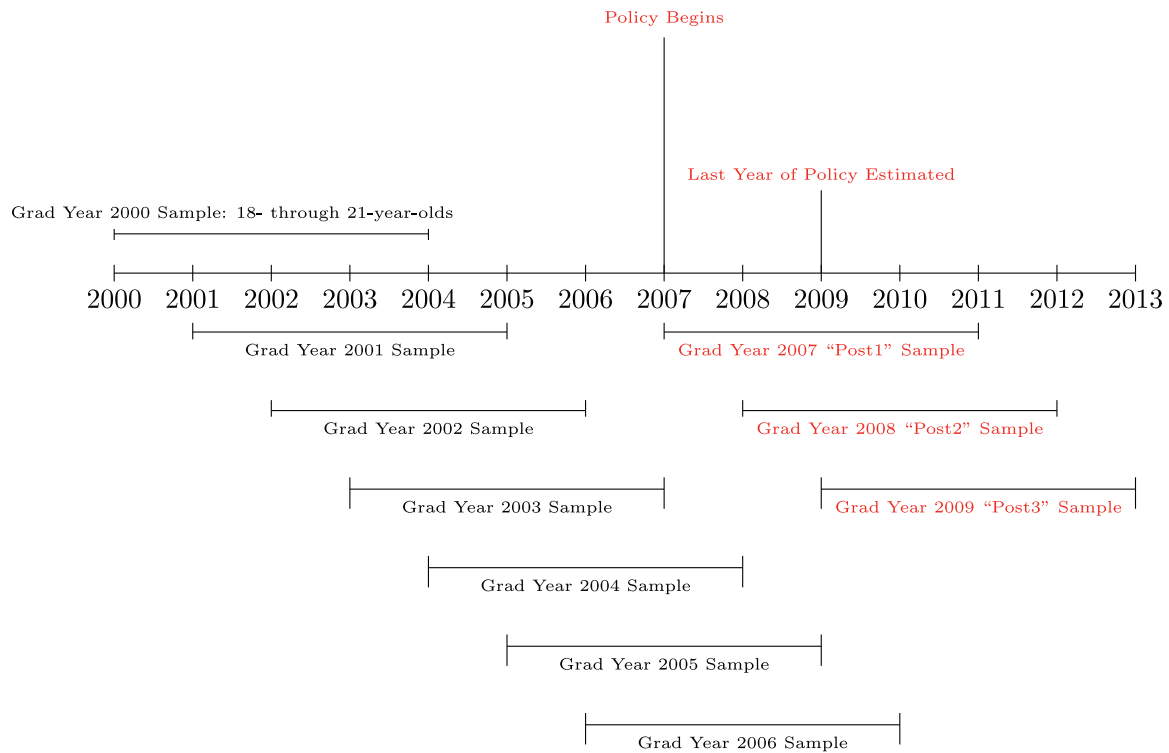


Fig. 2. Policy timeline and sample composition.

Table 2

Summary statistics, treatment versus synthetic control states for 2001 graduation cohort.

	GA	Control	All	TX	Control	All
Credit	607.6*	628.3	625.3	607.4*	627.9	620.7
Score	(88.39)	(82.70)	(83.87)	(86.8)	(83.1)	(85.0)
30 Days	0.0374*	0.0338	0.0344	0.0355	0.0348	0.0350
Behind	(0.190)	(0.181)	(0.182)	(0.185)	(0.183)	(0.184)
90 + Days	0.0501*	0.0438	0.0447	0.0549*	0.0453	0.0488
Behind	(0.218)	(0.205)	(0.207)	(0.223)	(0.208)	(0.215)
Unempl ⁺	4.83*	6.06	5.87	6.18*	6.41	6.33
Rate	(0.90)	(1.84)	(1.78)	(1.42)	(1.78)	(1.66)
# of	1.91*	2.24	2.19	2.142*	2.261	2.218
Accounts	(2.24)	(2.48)	(2.45)	(2.59)	(2.50)	(2.53)

Data: CCP 2001–2004. Notes: Means reported with standard deviations in parentheses. The data are for those who were 18 in 2001 as the first cohort that fully moves through the sample through age 21. * in the first column for each state indicates that we can reject the null of equality across the two groups using a t-test at the 0.05 level. Weighted controls are in Fig. 1.

later. To estimate this pattern, Table 3, adds indicators for the year in which the state enacted the policy and the subsequent two graduating cohorts. We represent the three years after the mandate was passed as θ_1 , θ_2 , and θ_3 . These results are particularly important because in Georgia and Texas, the mandates were passed three years before implementation. Our identification strategy in this section of the study relies on there being no effect of the mandate, and indeed, we see precisely estimated null effects of the policy passage. Two exceptions are in Columns (1) and (2) of Table 3, where we see a 0.49 percentage point decrease in 30 day defaults and a 1.1 percentage point reduction in 90+ day defaults in Georgia, though these are only statistically significant at the 10 percent level. Since we are testing 18 coefficients, we expect that roughly two may be statistically different from zero at the ten percent level. We take this as evidence that our parallel trends assumption is valid, while also highlighting that using the proper timing to estimate the effects is important.¹²

¹² When we include our post indicators β_1 , β_2 , and β_3 in this specification, neither our β nor our θ estimates substantively change.

Table 4 presents estimates of the effect of state personal financial education on credit outcomes for young adults in each of our treatment states. Panels A and B show the effect of the financial education graduation requirements in Georgia and Texas, respectively. Columns (1)–(3) include state and quarter-by-year fixed effects while Columns (4)–(6) include state-by-year and quarter fixed effects.

We begin in Column (1) of Table 4, where we that see across all states, financial education decreases the likelihood of being 30 days behind on an account. The effect sizes increase with each subsequent graduating class, and the magnitudes are comparable across states. For Georgia, the education reduced 30-day delinquencies by 1.84 percentage points in the third year of the mandate, which is roughly 50 percent of mean 30-day delinquency rates for the graduating class of 2001 (3.4 percent). Texas students saw a decrease of 30-day delinquencies of 1.4 percentage points, or about 40 percent of the mean rate (3.5 percent). These results remain consistent if we instead use state-by-year fixed effects in Column (4), though the estimates are larger in all specifications. The results from these three outcomes taken together may reflect that schools and teachers learn and adapt as they implement a

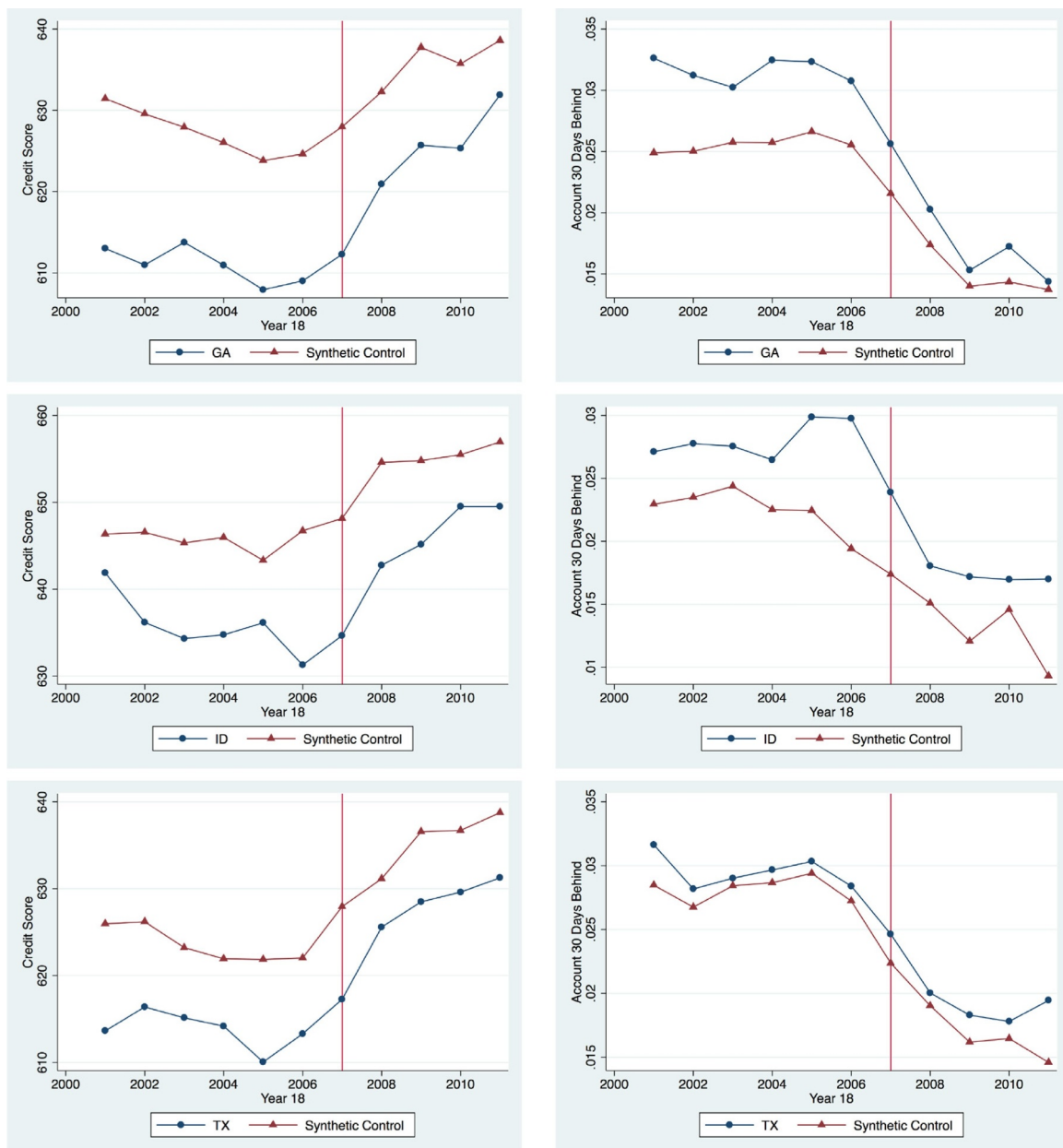


Fig. 3. Difference-in-differences plots: case study states implementing education mandates in 2007 compared to synthetic controls, by credit report Outcome. Data: CCP 2001–2014. Notes: The figures above show trends in average credit score (Column 1) and delinquency rates (Column 2) for each of the treatment states (Georgia and Texas) and their respective synthetic control weighted states. Column 2 shows the trend in 30-day delinquency; 90+ day delinquency plots are in Fig. 4.

new curriculum.

In both states, we find that financial education decreased the likelihood of being 90+ -days behind on an account. Like with 30-day delinquency, the education becomes more effective in its second and third year of implementation. In Georgia, the education reduced severe delinquencies by 1.3, 2.8, and 4.0 percentage points in years one, two, and three, respectively (Column (2)). With average 90+ -day defaults roughly 4.5 percent in 2001, this ranges from one-quarter, to one-half to nearly 90 percent of mean rates each consecutive year. In Texas, the results are similar in levels and averages to Georgia; years one, two, and three show a 1.7, 3.4, and 4.7 percentage point decrease in delinquencies, representing one-third, two-thirds, and nearly 95 percent reduction as a fraction of mean levels for each consecutive year.

In Column (3) of Table 4, we show the effect of each state's personal finance course requirement on credit scores. In Georgia, the first cohort

to receive the financial education is estimated to have a credit score 7.0 points greater than the comparison group, increasing to 18.3 points for the second cohort and 26.7 points for the third cohort. The magnitudes for the increases in credit score for Texas are similar to Georgia, with the first cohort having a credit score 5.7 points greater, increasing to 15.1 points for the second cohort and 23.1 points for the third cohort. To provide some context for the size of these estimates, Georgia and Texas average credit scores are 608 and 607, respectively. This means the effects vary slightly across states when comparing standard deviation units in Georgia (0.31 standard deviation units) and Texas (0.27 standard deviation units). When we instead incorporate state-by-year and quarter fixed effects, our estimates increase in magnitude and statistical significance across all specifications and graduation cohorts.

The implementation of state financial education mandates is associated with effects growing in magnitude for each subsequent cohort.

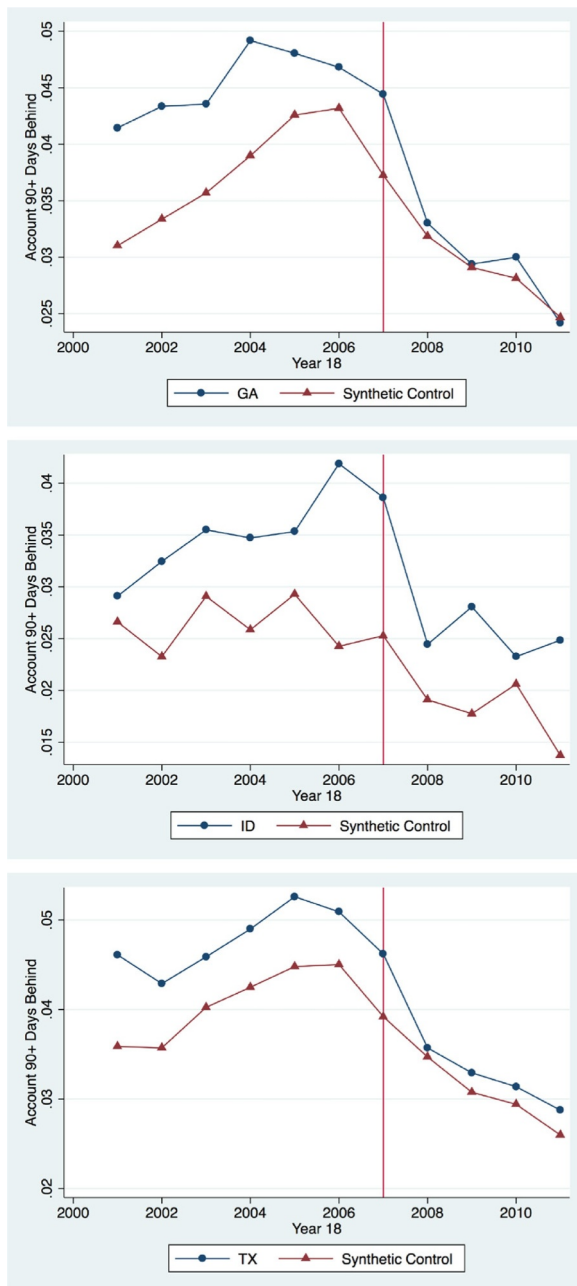


Fig. 4. Difference-in-differences plots: states implementing education mandates in 2007 compared to synthetic controls, by credit report outcome. Notes: The figures above show trends in average credit score and delinquency rates in the CCP data for each of the treatment states and their respective synthetic control states weighted. Trends for Georgia, followed by Texas, in 90+-day delinquency.

The increasing magnitude could be driven by delays in implementation across all schools within a state, thus reducing the number of students actually treated, or by teachers becoming more familiar with the course material over time and therefore better able to educate students. The relative difference may also be related to stronger peer and social network influences on credit behavior. Regardless, the general pattern is for later cohorts to perform better than the first cohort.

One concern that arises is that those exposed to rigorous financial education requirements are more (or less) likely to enter the credit report, changing the composition of the CCP sample. We assess the validity of this critique by testing the effects of financial education policies on people having a credit report and entering into the data. We

Table 3

Synthetic control estimates, by policy announcement year.

	(1) Account 30 Days Behind	(2) Account 90 + Days Behind	(3) Credit Score
Panel A: Georgia			
θ_1	−0.000880 (0.000972)	−0.000904 (0.00268)	1.818 (1.434)
θ_2	−0.00141 (0.00158)	−0.00434 (0.00396)	−0.732 (2.345)
θ_3	−0.00489* (0.00215)	−0.0113* (0.00525)	2.664 (3.319)
N	3,132,159	3,132,159	2,835,204
Panel B: Texas			
θ_1	−0.00151 (0.00187)	0.00286 (0.00668)	−6.366 (5.142)
θ_2	0.000809 (0.00284)	0.00513 (0.00947)	−10.70 (6.542)
θ_3	−0.00280 (0.00378)	−0.00529 (0.0120)	−5.584 (8.028)
N	3,388,035	3,388,035	3,039,025

Data: CCP 2001–2014. Notes: Robust standard errors clustered at the state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. θ_1 corresponds to the graduating class of the mandate passage; θ_2 corresponds to the following graduating class after the mandate; θ_3 is two classes after the mandate passes. $\theta_1 - \theta_3$ years all occur before the first graduating class is affected by the implementation. Models include state-level and quarter-by-year fixed effects, current county-level unemployment rate, state yearly operating education expense per capita, state annual support for higher education, and median household income by state and year. Models also include the interaction between each year post-implementation and the state as in Equation (2), though they are suppressed here for brevity. These regressions use the synthetic control weights.

pair aggregate counts of individuals in the CCP data by state, age, and year with population estimate data from the Census Bureau at the same unit of analysis. To address concerns about changes in the extensive margin, we show that financial education does not change the fraction of young people in a state's credit file. These results are in Table 5 Columns (1)–(2). Columns (3)–(4) further document that there are no clear pre-trends in credit score entry, though we do see two coefficients statistically different from zero (and one is close to zero in magnitude), in testing 10 coefficients, we expect that this can be due in part to chance. We also restrict the sample to individuals who appear in the data for at least three full years, or 12 quarters. These results, in Appendix Table A.2 are substantively similar to those in Table 4 and confirm that our findings are not driven by opening new credit.

We further supplement our main synthetic control estimates with an alternative specification sample using border states and contiguous border counties for both states. For each treatment state, we selected a bordering state that did not have a change in personal finance education requirements or a change in mathematics requirements. By focusing our comparison on individuals who are geographically more proximate, we hope to minimize unobserved heterogeneity that could bias the prior estimates. We explain this methodology in Appendix B, and show that the results are similar to those found using the synthetic control estimates in Table 4.

These results are generally supportive of graduation requirement policies. However, these two states implemented new financial education policies just as the Great Recession began. These new high school graduates faced a weak labor market, and may have entered into secondary education at a higher rate. While these were national trends that impacted all the students in the study in this time period, these estimates may be unique to this environment. The response of young people in a strong labor market could be different, although it is ambiguous whether results would be stronger or weaker.

Table 4
Case study samples: synthetic control estimates.

	(1) Account 30 Days Behind	(2) Account 90 + Days Behind	(3) Credit Score	(4) Account 30 Days Behind	(5) Account 90 + Days Behind	(6) Credit Score
Panel A: Georgia						
β_1	−0.00764*** (0.00106)	−0.0128*** (0.00287)	6.888** (2.182)	−0.00946*** (0.0000339)	−0.0172*** (0.0000357)	9.470*** (0.0627)
β_2	−0.0128*** (0.00134)	−0.0284*** (0.00364)	18.25*** (2.613)	−0.0151*** (0.0000749)	−0.0353*** (0.0000318)	22.32*** (0.0421)
β_3	−0.0184*** (0.00172)	−0.0404*** (0.00467)	26.73*** (3.180)	−0.0217*** (0.0000792)	−0.0468*** (0.0000414)	30.03*** (0.0529)
N	3,132,159	3,132,159	2,835,204	3,132,159	3,132,159	2,835,204
Panel B: Texas						
β_1	−0.00560* (0.00183)	−0.0172* (0.00584)	5.652 (4.834)	−0.00795*** (0.0000600)	−0.0246*** (0.0000596)	11.57*** (0.0542)
β_2	−0.0103** (0.00223)	−0.0343** (0.00696)	15.04* (5.510)	−0.0133*** (0.0000892)	−0.0437*** (0.0000865)	22.34*** (0.0790)
β_3	−0.0137** (0.00258)	−0.0468*** (0.00791)	23.00** (6.071)	−0.0166*** (0.000128)	−0.0556*** (0.000148)	29.74*** (0.116)
N	3,388,035	3,388,035	3,039,025	3,388,035	3,388,035	3,039,025
Fixed Effects Included						
State	Yes	Yes	Yes	No	No	No
Quarter-year	Yes	Yes	Yes	No	No	No
State-year	No	No	No	Yes	Yes	Yes
Quarter	No	No	No	Yes	Yes	Yes

Data: CCP 2001–2014. Notes: Robust standard errors clustered at the state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. β_1 corresponds to the first graduating class affected by the requirement; β_2 corresponds to the following graduating class; β_3 is two classes after the first graduating class. Models include current county-level unemployment rate, state yearly operating education expense per capita, state annual support for higher education, and median household income by state and year. Columns (1)–(3) include state and quarter-by-year fixed effects. Columns (4)–(6) include state-by-year fixed effects as well as quarter fixed effects. These regressions use the synthetic control weights.

6. Conclusions

The political economy of state education policy is complex. The trend toward offering financial education in school is based on the belief that additional financial education has the potential to provide students with financial knowledge that will help them develop future positive financial behaviors. This study suggests that, at least in these two states rigorously implementing a personal finance graduation requirement, financial education has positive effects on the credit behaviors of young adults. This study also illustrates the importance of understanding policy heterogeneity when estimating the effects of education programs on observational outcomes.

Our work builds upon the findings of two recent studies. First, [Cole et al. \(2015\)](#) find that personal finance education does not change financial behaviors for the middle-aged. Second, [Brown et al. \(2016\)](#) show that financial literacy requirements improve credit scores and lower delinquency rates for those up to age 29. Our work suggests that the strategy in both [Cole et al. \(2015\)](#) and [Brown et al. \(2016\)](#) to indicate that states either do or do not have financial education mandates masks important heterogeneity in policy. We show some heterogeneity in effect sizes even within two states that implemented comparable graduation course requirements in the same time frame.

We further build upon [Cole et al. \(2015\)](#) and [Brown et al. \(2016\)](#) by showing that the timing of the policy used in the evaluation matters. If we use the approval years instead of the policy implementation year, our estimates are no longer significant. Finally, we suggest that studying individuals just as they exit school is important. If financial education policies do not affect students as they establish their financial independence, it is unlikely to have effects later in life. However, it is difficult to empirically document long-run effects of the policy, as these effects may be confounded with many other life changes, such as marriage, family formation, labor market shocks, and other demographic and economic factors. Overlapping policies and defining standard pre- and post-periods across states and graduation cohorts also makes longer-run estimates less precise.

The experiences of Georgia and Texas are instructive for education policy. Both states had some level of curriculum coordination, supplemental teaching resources, teacher training, and certification requirements. However, Georgia's program had extensive and continuous teacher training, additional certification incentives, and student testing. Georgia also had the larger improvements in credit outcomes. Georgia's enhanced implementation efforts may be part of the reason the state's financial education program appears to have relatively stronger effects on young people.¹³

As a stylized illustration of the magnitude of our effects, consider that about 265,000 students graduated from high school in Texas in 2007, and we would expect 200,000 to have active credit files by 2011. On average, 4.8 percent of young adults with active credit records have a 90-day late payment by age 22, implying that about 10,000 of these young people in Texas would experience a default. The estimates in [Table 4](#) would predict that more than 5,000 of these young people in the class of 2007 would have avoided a serious credit default after the education was in place. Given the higher costs of credit associated with a history of serious default, these students are likely better off, at least in terms of lower interest rates and fees and expanded credit access. The state's policy allows affected students to consume more of their income in future periods and spend less on borrowing costs. These policies may potentially be a benefit for lenders as well, by allowing them to avoid collections and loan write-off costs. To the extent that defaults have negative externalities for families and communities, the implementation of high school financial education requirements may allow students and the wider community to avoid other costs, as well.

Of course, there are opportunity costs for students and schools associated with personal finance graduation requirements, and a more robust math, art, or other requirement might also yield positive benefits for young people. We cannot assess the relative merit of a personal

¹³ We cannot rule out that there was heterogeneity in effect sizes based on the student-characteristics within the state, however, only that differences exist.

Table 5
Synthetic control state sample extensive margin results.

Dependent Variable = Fraction with Credit File				
	GA (1)	TX (2)	GA (3)	TX (4)
β_1	−0.0031 (0.0048)	0.0068 (0.0044)	0.0005 (0.0069)	0.0081 (0.0059)
β_2	−0.015** (0.0044)	0.0086 (0.0078)	−0.0114 (0.0066)	0.0099 (0.0078)
β_3	−0.00094 (0.0059)	0.0069 (0.0034)	0.0026 (0.0021)	0.0082 (0.0051)
$\hat{\beta}_2$			0.0188*** (0.0029)	0.0062** (0.0012)
$\hat{\beta}_3$			0.0147 (0.0067)	0.0028 (0.0038)
$\hat{\beta}_4$			−0.0015 (0.0072)	−0.0005 (0.0089)
$\hat{\beta}_5$			0.0062 (0.0079)	0.0021 (0.0073)
$\hat{\beta}_6$			−0.0133 (0.0089)	−0.0015 (0.010)
N	348	172	348	172
Mean DV	0.481	0.467	0.481	0.467

Data: CCP 2001–2014. Notes: Robust standard errors clustered at the state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable equals the fraction in the sample in the credit file divided by the Census Bureau's population estimate for the same age range. β_1 corresponds to the first graduating class affected by the requirement; β_2 corresponds to the following graduating class; β_3 is two classes after the first graduating class; $\hat{\beta}_2 - \hat{\beta}_6$ correspond to the graduating classes 2 through 6 years before the mandate. The excluded group is the graduating class the year before the mandate took place. Models include state, year, and age fixed effects. These regressions use the synthetic control weights.

finance mandate versus other mandates. We also must remain circumspect regarding how long young people will sustain these positive credit habits. More financial education could simply end up delaying problematic behavior. It is possible that making mistakes in managing credit helps people to manage credit better later in life than a formal high school course. Future studies could ideally better measure how behaviors are influenced by financial education over time, especially as young people make major personal financial decisions related to housing and other matters.

Based on our results from Georgia, well-funded teacher preparation may be important for successful financial education programs. This presents a direct cost for school systems, however. If there are 500 teachers in a state providing financial education courses, then even a \$500 training per teacher could be a significant cost for a state (\$250,000). However, not all teachers need training every year, and the costs might reasonably be spread over several years. Even if a portion of Georgia's relatively large impacts from financial education can be attributed to support for teachers, it may be a worthwhile investment.

Overall, the findings of this analysis are promising. To the extent that shifting the credit trajectory of young adults is a policy goal, well-implemented financial education mandates may prove to be an effective mechanism. Beyond the context of financial education, this study is also instructive for policy analysts in general. High-quality policy analysis, especially when studying policies that are implemented at the state and local level, requires a deep understanding of the details of how and when programs are actually put into place. Field studies and observational studies often can provide useful estimates of the causal effects of policies, but only if the indicators of these policies are defined accurately. Often policy implementation is complicated and messy—what policymakers pronounce may or may not be what actually occurs. Researchers need to be well informed on these issues, and exercise caution when using dichotomous indicators of policy vs. no policy populations or jurisdictions. As the saying goes, “the devil is in the details.” This study shows that how details of policy implementation are classified and used in analysis can result in widely different estimates of impacts.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.econedurev.2018.03.006](https://doi.org/10.1016/j.econedurev.2018.03.006).

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